GEK-88205

DRIVE SYSTEM INSTRUCTIONS

AYDIN MONITOR

INSTALLATION AND MAINTENANCE MANUAL

DRIVE SYSTEMS DEPARTMENT
SALEM, VA. 24153

GENERAL ELECTRIC

NOTICE

This manual is a reproduction of selected portions of the Aydin Controls "OPERATION AND MAINTENANCE MANUAL". Complete copies of the manual can be obtained by ordering document no. 150-7030-001 from:

AYDIN CONTROLS
414 Commerce Drive
Fort Washington, PA 19034

NOTICE

The information herein does not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation, and maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company, Drive Systems Operations, 1501 Roanoke Boulevard, Salem, Virginia, U.S.A., 24153.

TABLE OF CONTENTS

Section	Title	Page
1	GENERAL DESCRIPTION	
	1.1 Introduction	1
	1.2 Physical Description	1
	1.3 Characteristics	1
	1.4 Recommended Test Equipment and Special Tools	1
2	INSTALLATION	
	2.1 General	7
	2.2 Unpacking	7
	2.3 Mounting Instructions	7
	2.4 Video Inputs	7
	2.5 AC Input Power	11
	2.6 Remote Health Check Indicator	11
	2.7 Installation Checkout	11
3	OPERATION	
	3.1 Introduction	12
	3.2 Preoperational Setup	12
	3.3 Operation	12
	3.3.1 Front Panel Controls	12
	3.3.2 Drawer Controls	13
4	MAINTENANCE	
	4.1 General	14
	4.2 Preventive Maintenance	14
	4.3 Alignment Procedures	14
	4.3.1 Degaussing	15
	4.3.2 D.C. Power Supply Adjustments	15
	4.3.3 High Voltage Adjustment	16
	4.3.4 Horizontal Adjustments	16
	4.3.5 Vertical Adjustments	17
	4.3.6 Underscan Mode	18
	4.3.7 Purity Alignment	19
	4.3.8 Static Convergence Procedures	20
	4.3.9 Pincushion Correction Circuit (PCC)	20
	4.3.10 Screen Bias and Video Drive	21
	4.3.11 Focus	21
	4.3.12 Dynamic Convergence Alignment Procedure	22
	4.3.13 Video Circuits	22
	4.3.14 Automatic Beam Limiter (ABL) Adjustment	23
5	PARTS	24

LIST OF FIGURES

Figure	Title	Page
1-1	8025 CRT Color Monitor PCBs and Major Components	2
2-1	Cabinet Style 8025 CRT Monitor	8
2-2	Rack Mounted 8025 CRT Monitor	9

LIST OF TABLES

Table	Title	Page
1-1	8025 CRT Monitor Characteristics	3
2-1	Rear Panel Connectors	10
5-1	Spare Subassemblies and Chassis Piece Parts	24

SECTION 1

GENERAL DESCRIPTION

1.1 INTRODUCTION

This manual contains installation and maintenance information for the Aydin Controls 8025 CRT Color Monitor. The monitor is an ultra high resolution red, green, blue (RGB), shadow mask display device designed for compatibility with alphanumeric and graphic raster-scan display systems. It features a fine pitch shadow mask CRT and separate red, green, and blue video channels to meet high resolution display requirements. The unit also has provision for external synchronization.

To assure compatibility with standard industrial TV design techniques, the monitor meets the specifications of EIA Standard RS-170. It also complies with DHEW PHS Rules and Regulations, Part 78, relating to X-ray radiation.

1.2 PHYSICAL DESCRIPTION

The 8025 CRT Color Monitor has a finished chassis cabinet enclosure designed for table top application. For rack-mounted applications, the cabinet enclosure should be removed and slides attached to the side of the chassis. Mounting holes and hardware are provided for slide mounting.

The location of the printed circuit boards (PCBs) and major components within the 8025 CRT Monitor are listed and depicted in Figure 1-1. Replacement parts are included in Section 5.

1.3 CHARACTERISTICS

The essential electrical performance and physical characteristics of the 8025 CRT Color Monitor are listed in Table 1-1.

1.4 RECOMMENDED TEST EQUIPMENT AND SPECIAL TOOLS

The following test equipment and special tools, or their equivalents, are recommended for maintaining the 8025 CRT Color Monitor:

a. Digital Voltmeter, Weston Model 1240.

b. High Voltage Probe, 30KV - Simpson, Catalog Number 00509.

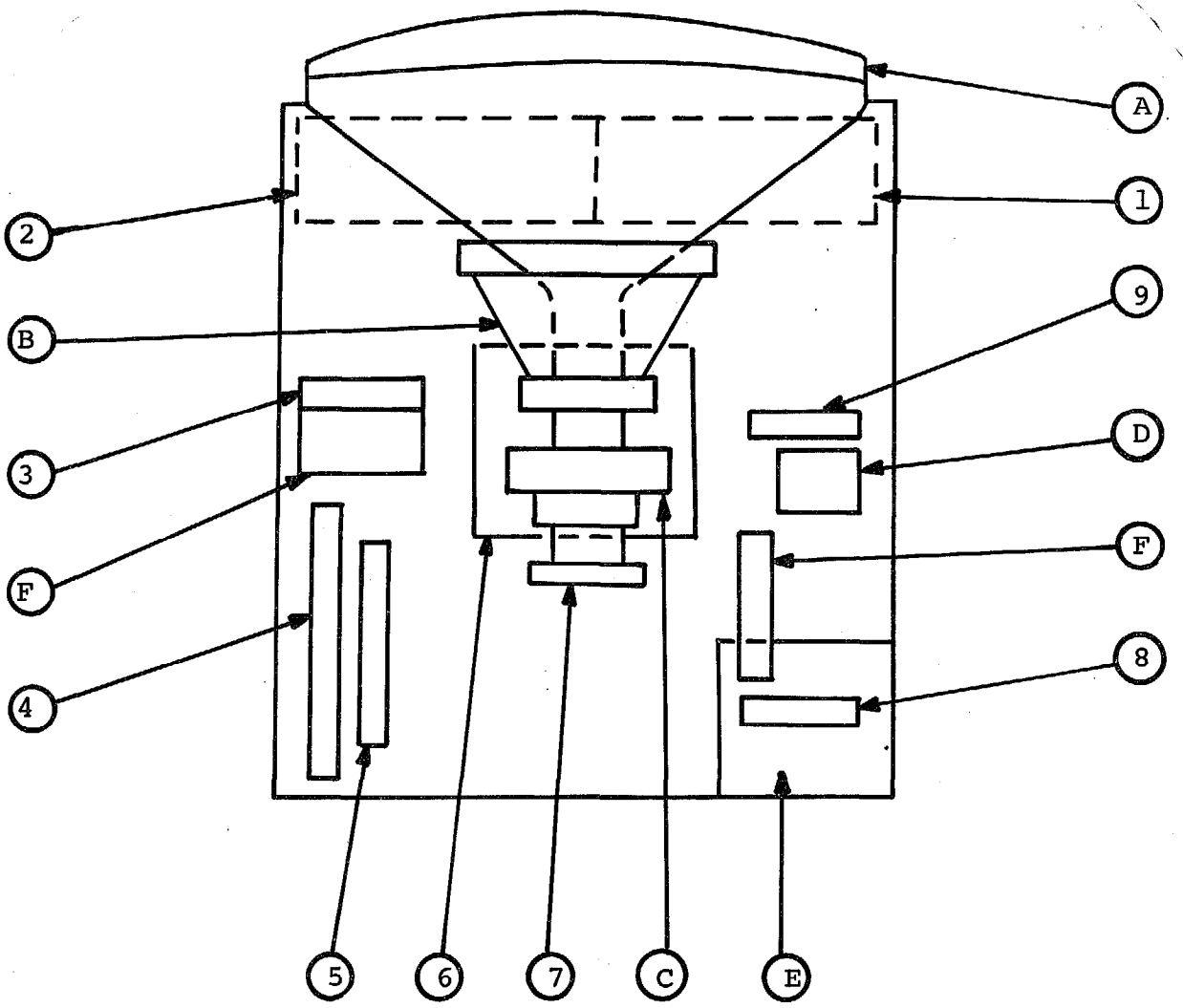
c. Oscilloscope - Hewlett Packard, Type 184A, 2-Channel, 50 MHz; Amplifier, Type 1801-A; Time Base and Delay Generator, Type 1821A; Two 10-1 probes, 10004D; and a 50-to-1 probe, 10002A.

d. Degaussing Coil - G. C. Electronics, Catalog Number 9317.

e. Hex Tuning Tool - G.C. Electronics, Catalog Number 8284.

f. Alignment Screwdriver - G.C. Electronics, Catalog Number 5009.

g. Duplex Steel Blade Aligner - G.C. Electronics, Catalog Number 8277.



1. PCB - Convergence
2. PCB - Control
3. PCB - Power Regulator
4. PCB - Video
5. PCB - Power Rectifier
6. PCB - Connector
7. PCB - CRT
8. PCB - Deflection
9. PCB - High Voltage

A. CRT
B. Deflection Yoke
C. Convergence Yoke
D. High Voltage Transformer
E. Power Transformer
F. Heat Sink

Figure 1-1. 8025 CRT Monitor PCBs and Major components

Table 1-1. 8025 CRT Monitor Characteristics

Characteristics	Description
ELECTRICAL	
o Video (RS 170 Compatible)	Composite: Loop-through or switchable to 75 ohms termination; 1.0V P-P nominal (0.35V to 2.0V) sync negative. Non-Composite: Loop through or switchable to 75 ohms termination; 0.7V P-P nominal (0.25V to 1.4V) black negative
o Data Input	Separate inputs: red, green, and blue; composite on green for synchronization or external sync
o External Sync	Composite Sync: Loop-through or switchable to 75 ohms termination; 2.0V P-P nominal (1.0V to 4.0V) sync negative
o Input Connector	BNC type
o Video Bandwidth	± 3db, 50 Hz to 15 MHz for each input: red, green and blue
o Video Amplifier Differential Gain	Less than 5% for luminance range of 0 to 20 FL
o Horizontal Frequency	Adjustable over a range of 15 to 18 kHz
o Horizontal Retrace Time	Less than 8 us.
o Vertical Retrace Time	Less than 600 us.
o Input Impedance	Loop-through or switchable to 75 ohms termination on red, green, blue and external sync
o Input Power	120VAC ± 10%, 50/60 Hz; less than 250VA; 100V, 110V, 220V, 240V ± 10% operation selected by internal connector change

Table 1-1. 8025 CRT Monitor Characteristics (continued)

Characteristics	Description
ELECTRICAL (continued)	
o Electrical Components	100% solid-state with the exception of CRT
o Display Area	Effective display area 15.25 in. wide by 11 in. high (nominal)
o CRT	19-inch diagonal, high resolution, 3-gun shadow mask CRT
o Temperature Operation	-5° to +40°C
PERFORMANCE	
o Convergence	Does not deviate more than 0.5 mm in a centrally located area bounded by the circle with a diameter equal to the picture height. Elsewhere does not deviate more than 1.0 mm
o Linearity and Geometry	No point on the raster deviates from its proper position by more than 2% of raster height
o Raster Regulation	Less than 1% change in size; 0% to 100% APL (Average Picture Level) at 20 FL peak luminance
o Phosphor Resolution	Over 900 TV lines. Ultimate resolution limited by the CRT which has approximately 1320 triads horizontally and 1640 triads vertically

Table 1-1. 8025 CRT Monitor Characteristics (continued)

Characteristics	Description
MECHANICAL	
o Size - Cabinet Model	19 in. wide, 18 in. high, 21.5 in. deep
- Rack Model	19 in. wide, 17.5 in. high, 19.8 in. deep (Within Rack)
o Weight	100 lbs. with cabinet
o Construction	All metal chassis with glass epoxy printed circuit cards
CONTROLS	
o Front Panel	Brightness - potentiometer
	Power - 2-position switch
	Degauss - momentary switch
	Color/Monochrome - slide switch
	Power On/Health Indicator - LED
o Drawer	Convergence controls - 10 groups of 3 each (RGB) for 10 areas of the screen
	Gain
	Height
	Width
	Horizontal hold
	Vertical hold
	Linearity
	Pincushion controls
	Dynamic focus
	RGB Beam on/off switch
	RGB Background - potentiometers
	RGB Contrast - potentiometers
	Scan/Underscan Switch

Table 1-1. 8025 CRT Monitor Characteristics (continued)

Characteristics	Description
CONTROLS (continued)	
o Rear Panel	Red Input - 2 coaxial connectors
	Green Input - 2 coaxial connectors
	Blue Input - 2 coaxial connectors
	Ext. Sync - 2 coaxial connectors
	Red - 75 ohm termination switch
	Green - 75 ohm termination switch
	Blue - 75 ohm termination switch
	Ext Sync - 75 ohm termination switch
	Remote Health Check Connector
	Power Connector

SECTION 2

INSTALLATION

2.1 GENERAL

This section contains procedures for installing and verifying the basic operation of the 8025 CRT Color Monitor. The monitor is thoroughly adjusted and checked out at the factory, but it may require certain minor adjustments to compensate for minor adjustment disturbances caused during shipment. Complete adjustment procedures and other basic checks are described in Section 4; however, only selected, simple procedures should be necessary for initial installation.

2.2 UNPACKING

The 8025 CRT Color Monitor, which is shipped completely assembled, is normally packaged in a separate shipping container. Unpack the monitor as follows:

a. Carefully open top of container.

b. Remove inside packing material and lift out monitor.

2.3 MOUNTING INSTRUCTIONS

The cabinet style monitor (Figure 2-1) is simply placed on a supporting surface within the user's system configuration. For rack-mounting, proceed as follows:

a. Secure slide receptacles (rails) to equipment rack. (Refer to Figure 2-2 for mounting details.)

NOTE

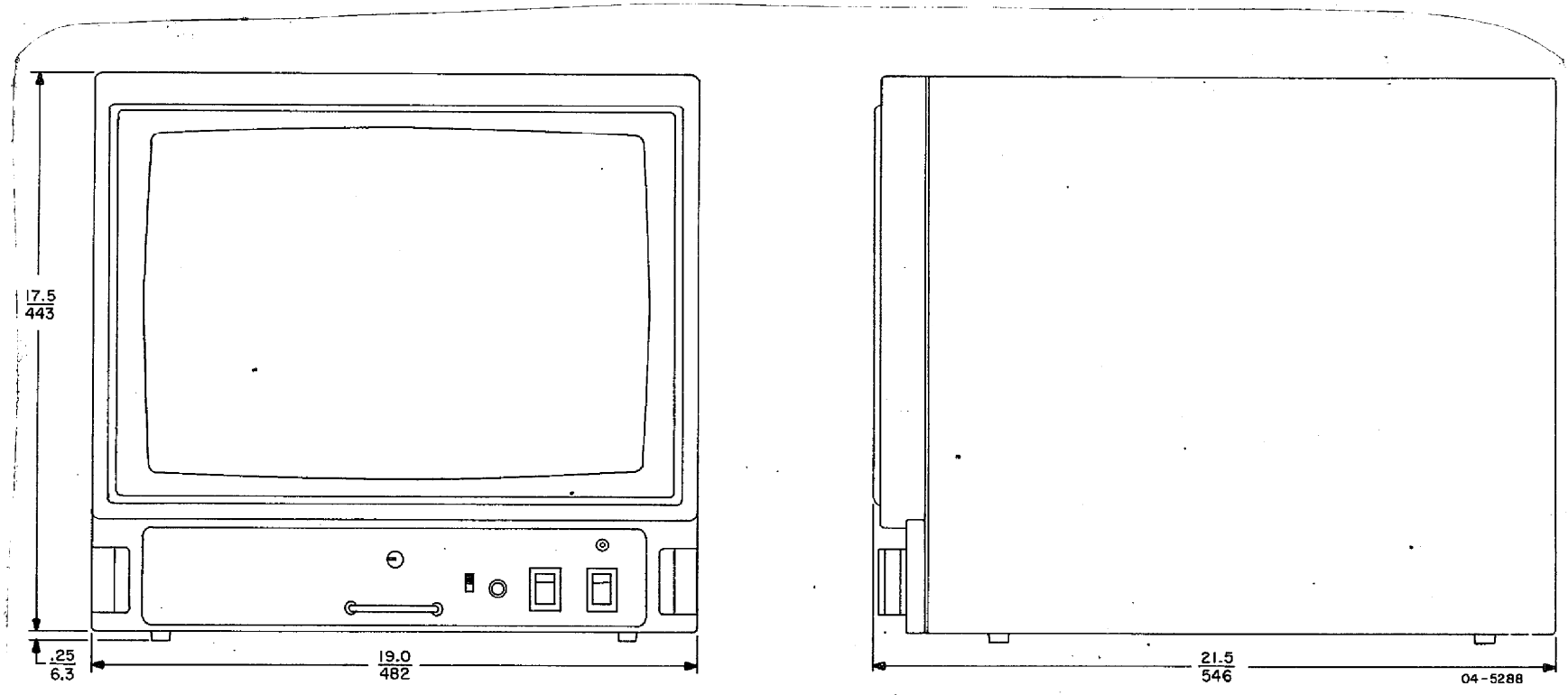
The 8025 CRT Color Monitor is designed for slide mounting in standard 19-inch equipment racks. Hardware is provided.

b. Mate chassis slides on monitor with slide receptacles in the cabinet and slide monitor into position.

c. Fasten retaining screws to secure monitor in place.

2.4 VIDEO INPUTS

The red, green, and blue video cables should be connected to the appropriate video input connectors located on the rear connector panel. Refer to Table 2-1.



Front View

Left Side View

Figure 2-1. Cabinet Style 8025 CRT Monitor

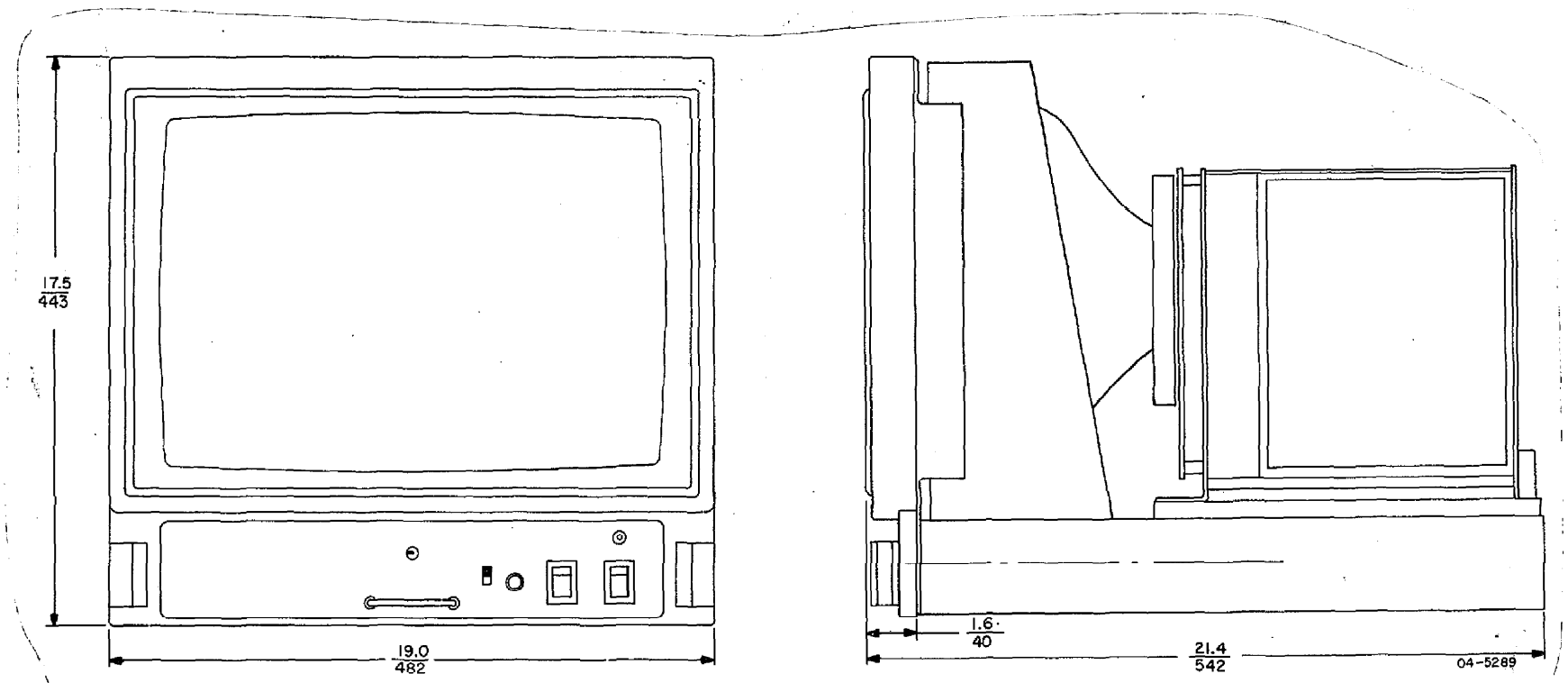


Figure 2-2. Rack Mounted 8025 CRT Monitor

NOTE

The video cables should be constructed with 75 ohm coaxial cable (Type RG59/U, or equivalent) and terminated, at the monitor cable end, with standard BNC connector plugs. No separate sync input is required. The green video signal input to the monitor must contain composite sync. An external sync connector is provided for external sync control if desired.

Table 2-1. Rear Panel Connectors

Connector	Function	Connector Type
J191	Blue Video Input	BNC
J192	Blue Video Output	BNC
J193	Red Video Input	BNC
J194	Red Video Output	BNC
J195	Green Video with Composite Sync Input	BNC
J196	Green Video with Composite Sync Output	BNC
J197	Ext. Sync Input	BNC
J198	Ext. Sync Output	BNC
	Remote Health Check Indicator	
	120 VAC ± 10% Power Input 50/60 Hz	

The 8025 CRT Color Monitor contains two connectors for each video input in order to provide for loop-through operation. In addition, the input impedance of each input may be set to a high impedance or to 75 ohms by means of three input impedance selector switches, located below the video input connectors. If the monitor is used in a loop-through operation, these switches should be set to the high impedance position. For single unit operation, or when the monitor is the last unit in a loop-through string, these switches should be set to the 75 ohm position to terminate the video cable.

2.5 AC INPUT POWER

Before connecting the monitor to the power source, perform the following:

a. Determine that line voltage and frequency are proper (120 VAC ± 10%, 50/60 Hz).

b. Check position of input voltage plug on Power Rectifier card.

c. If line voltage being used is not 120VAC, make certain the plug is in the proper position for input line voltage desired (100V, 110V, 220V, 240V).

d. Plug ac line cord into power receptacle on rear connector panel.

2.6 REMOTE HEALTH CHECK INDICATOR

On the rear panel, a 4-pin connector is provided for a remote health check indicator, which provides the same function as the front panel health check indicator. The output circuit is designed to provide approximately 12 ma to an external LED indicator.

2.7 INSTALLATION CHECKOUT

Proceed with installation checkout as follows:

a. Apply a test pattern or some other suitable video signal to video inputs.

b. Connect monitor to an ac power source and set power switch to ON position. (There is normally a 30-second delay from the application of power to the appearance of an image.)

c. Determine that the health check indicator illuminates, indicating a proper and safe high voltage level, low voltage, and sweep circuit operation.

d. Adjust brightness and video gain for desired viewing level.

e. If necessary, adjust the sync and contrast controls located in the drawer, and the focus control located on the HV assembly.

f. If color purity or convergence are slightly out of adjustment, press degauss control, hold for 5 seconds and release.

g. If monitor does not display an image, or if displayed image is maladjusted and cannot be corrected, refer to maintenance and alignment procedures in Section 4.

SECTION 3

OPERATION

3.1 INTRODUCTION

This section describes the procedures that should be followed in operating the 8025 CRT Color Monitor.

3.2 PREOPERATIONAL SETUP

Before operating the monitor, perform the following procedures:

a. Apply video input signals to the monitor as specified in the installation procedures of Section 2.

b. Determine that the input voltage selection plug on Power Rectifier card is in proper position for the applied line voltage (refer to paragraph 2.5).

c. If it is desired to change the line rate of the monitor, adjust the Horizontal Hold control until monitor synchronizes and a stable picture results. This monitor will synchronize between 15 kHz and 18 kHz.

NOTE

For additional adjustments, refer to Section 4.

3.3 OPERATION

3.3.1 Front Panel Controls

a. Apply ac power to the monitor and press ac power switch to turn on power.

b. Observe that health check indicator illuminates, indicating a proper and safe high voltage, low voltage, and sweep circuit operation.

c. Allow approximately 30 seconds for warmup.

d. Adjust the brightness control until the raster is barely visible and then adjust the video gain (located in drawer) until the desired viewing level is achieved. Use brightness control to adjust level after initial adjustment.

3.3.1 Front Panel Controls (continued)

e. If the color purity or the convergence appear to be slightly out of adjustment, press the degauss switch and hold it for 5 seconds before releasing.

f. If display image is not satisfactory, refer to purity and convergence alignment procedures in Section 4.

NOTE

The Color/Mono switch may be used to produce a monochrome display.

3.3.2 Drawer Controls

a. If the raster is not synchronized, adjust horizontal hold and vertical hold controls to obtain proper sync.

b. The three color contrast controls are adjusted to obtain white highlights. The three color switches are provided to eliminate any color and to provide combinations which will facilitate convergence of the monitor.

NOTE

For any additional adjustments refer to Section 4.

SECTION 4

MAINTENANCE

4.1 GENERAL

This section explains how to maintain the Aydin Controls 8025 CRT Color Monitor. Maintenance techniques are presented for preventive and corrective procedures. The procedures are supported by voltage, waveform, and alignment information. Refer to paragraph 1.4 for a list of recommended test equipment and tools.

```
**********
*        *
* CAUTION *
*        *
**********
```

Be extremely careful when making voltage or waveform checks on a live chassis because of dangerously high internal voltages. Completely disconnect the monitor from its ac source before changing cards and chassis components.

4.2 PREVENTIVE MAINTENANCE

Preventive maintenance for the monitor consists primarily of basic housekeeping tasks. Check all connections for proper fit, corrosion, sharply bent cables, exposed wires and broken plugs. Periodically turn off all power and air clean the chassis interior. Wipe off the chassis exterior and the CRT face with a lint-free cloth dampened with a mild cleaning solution. The cleaning frequency is determined by the environmental control at the user's site.

4.3 ALIGNMENT PROCEDURES

Monitor alignment procedures contained in this section should be followed after the replacement of a major component, such as a CRT, deflection yoke, convergence assembly, or circuit board. Some alignment may also be required periodically to correct for component aging. Convergence and purity alignment may also be required following installation or relocation of the monitor. Degaussing should be performed periodically whenever it is suspected that degaussing is required. These alignment procedures should be performed in the order given herein. Due to interaction, some portions of the alignment procedures may require repeating.

In the following alignment procedures it is assumed that proper line voltage and frequency are available. A video source with proper line rate is required for application to the red, green, and blue inputs. The green video must contain composite sync or a proper signal must be supplied to the external sync input. The video source should be a cross-hatch or dot pattern for convergence alignment. After all inputs have been connected, the horizontal hold (VR571) and the vertical hold (VR471) must be adjusted for a stable picture. Approximately 30 minutes should be allowed for warm-up before proceeding.

4.3.1 Degaussing

The need for degaussing is indicated by poor color purity or poor convergence. Degaussing should be performed before any color purity or convergence alignment is attempted. The display monitor is equipped with a built-in picture tube degaussing coil. To degauss the picture tube the degauss switch S992 on the front panel is pressed and held for approximately 10 seconds to ensure proper operation of the degauss circuit.

Other parts of the monitor may also require degaussing. This would be indicated by poor color purity or convergence which cannot be corrected by normal alignment. Degaussing of the monitor chassis is performed manually by using a commercial degaussing coil. The following procedure should be adhered to when using a degaussing coil:

a. With the coil switch in the OFF position and the degaussing coil 6 to 8 feet from, and perpendicular to the screen, turn the switch to the ON position.

b. Turn the coil parallel to the screen and, with a circular motion, slowly bring the coil to the monitor.

c. Continuing the circular motion, pass the coil over the front, top, and sides of the monitor for approximately two minutes.

d. Then, moving in a circular motion and with the coil perpendicular to the monitor, slowly back away 6 to 8 feet and turn the coil switch OFF.

4.3.2 D.C. Power Supply Adjustment

4.3.2.1 To obtain 130VDC proceed as follows:

a. Connect a dc voltmeter, set to measure +130VDC, to test point TP91 on Power Regulator PCB and the chassis.

b. Apply ac line power to the monitor and set front panel power switch S991 to ON position.

c. Adjust potentiometer VR931 to obtain +130 $\pm$ 1VDC.

4.3.2.2 To obtain 100VDC, proceed as follows:

a. Connect a dc voltmeter, set to measure +100VDC, to test point TP92 on Power Regulator PCB and the chassis.

b. Apply ac line power to the monitor and set front panel power switch S991 to the ON position.

c. Adjust potentiometer VR932 to obtain +100 $\pm$ 1VDC.

4.3.1.2.3 For +60VDC, proceed as follows:

a. Connect a dc voltmeter, set to measure +60VDC, to test point TP93 on Power Regulator PCB and the chassis.

b. Apply ac line power to the monitor and set front panel power switch S991 to the ON position.

c. Adjust potentiometer VR933 to obtain +60VDC ± 1VDC.

4.3.3 High Voltage Adjustment

Before attempting any high voltage measurements, perform the following:

a. Disconnect the unit from the ac line source.

b. Take a grounding strap, connect one end to the chassis and push the other end under the rubber anode cap on the side of the CRT.

c. Repeat this operation several times to make sure there is no residual charge at this point.

To adjust high voltage, proceed as follows:

a. With power switch S991 in the OFF position, connect a voltohmmeter with a high voltage probe between the anode cap of the CRT and the chassis.

b. Turn potentiometer VR301, located on the High Voltage PCB, fully counterclockwise.

c. Set power switch S991 to the ON position and gradually turn VR301 clockwise until a reading of 25KV ± 100V is achieved.

d. Turn unit off before removing the meter connections from the anode cap of the CRT.

4.3.4 Horizontal Adjustments

Before the horizontal alignments are performed, proper line voltage should be applied and the power supply dc output adjustments must be proper.

4.3.4.1 Horizontal Hold

To adjust the horizontal hold, proceed as follows:

a. Apply a cross-hatch pattern to the red, green, and blue video inputs. Use either composite sync on the green input or connect proper sync through external sync input.

b. Adjust horizontal hold control VR571, located inside the drawer, so that it matches the proper line rate being applied to the monitor.

NOTE

Proper line rate is achieved when the raster appears without tearing. The horizontal hold control will synchronize raster within the horizontal frequency range of 15KHz to 18KHz.

ITEM	QUAN.	APPARATUS

DC Voltage Capability
100% rated DC voltage will be maintained at 180% rated current with 90% rated AC voltage at the terminals of the converter.

1 - Lot of control equipment including:

- Current limit - adjustable
- Speed regulator (1% of top speed accuracy)
- Current sensing permissive for switching
- Speed set by separately mounted operator potentiometer in manual mode of operation and by isolated remote 0-10 volt signal in PMS mode of operation.
- Sequencing for contactors

Controls on Door of Enclosure

- Lights for drive running, adjustable speed range, and maximum fixed speed.
- DC ammeter
- DC voltmeter
- Speed indicator

1 - Speed potentiometer (manual 0-10 volt reference)
1 - SB1 switch for local/PMS operation, key-locked
1 - Start push button.
1 - Stop push button.

1D 2 Reactor, dry type air cooled, in NEMA 1 enclosure for connection between diode rectifier and thyristor power converter. Inductance = 2.6 mH.

1F 2 Converter Transformer.

Indoor, self ventilated, dry type, rated 150°C rise continuous for ANSI (T1) duty class, rated (417) KVA, primary 4160V. secondary rated (487) volts, (60) Hz. Impedance 0.1 PU.

Rating based on maximum ambient temperature (40°C).

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4.3.5.2 Vertical Centering

Three selections of vertical centering are provided. These selections are made by changing the position of vertical centering switch SW401. The position of the raster should be observed in order to determine how to adjust SW401. The following are the adjustments possible:

a. Set SW401 to upper position to move the raster closer to the bottom of the CRT.

b. Set SW401 in center position to maintain the picture in a position midway between the upper and lower positions of the CRT.

c. Place SW401 in lower position to move the raster closer to the top of the CRT.

NOTE

If SW401 is in the upper or lower position, VR404 provides a means of varying the raster to the desired final position.

4.3.5.3 Vertical Height and Linearity

To adjust vertical height and linearity, proceed as follows:

a. Apply video containing a cross-hatch test pattern to the monitor RGB inputs. Synchronize the raster, if required.

b. Adjust front panel brightness control VR691 for suitable brightness.

c. Position switch SW971 to the left on the PCB control board to select the scan mode.

d. Adjust the height control VR401, located on the PCB Control, to obtain a raster height of 90 percent of full screen.

e. Adjust vertical linearity control VR403, located on PCB control, for best vertical symmetry.

f. Readjust the height and linearity controls, if required, by repeating steps d and e.

4.3.6 Underscan Mode

The monitor is capable of displaying an overscanned raster approximately 3 to 5 percent or an underscanned raster approximately 10 percent. When changing to the scan mode, the height and width should be adjusted according to paragraphs 4.3.4.3 and 4.3.5.3.

When the underscan mode is used, the following adjustments are necessary:

a. Apply video containing a cross-hatch test pattern to the monitor RGB inputs.

b. Position switch SW971 to the right.

c. Adjust brightness control VR691 for suitable brightness.

d. Adjust height control VR402 and the width control VR972, located on the control card, so that the four corners of the raster are visible.

NOTE

If vertical linearity requires adjustment, refer to paragraph 4.3.5.3.

4.3.7 Purity Alignment

Purity alignment is performed using the following procedure:

a. Before a purity alignment is attempted, the monitor should be degaussed. Refer to the degaussing procedure in paragraph 4.3.1.

b. Place the monitor in the position that it will be used or orient the monitor to face north or south.

c. Apply a cross-hatch composite video signal to each video input. Operate the monitor for at least 30 minutes before proceeding.

d. Observe the display to determine that the raster center is converged and to determine that the dynamic convergence is approximately correct.

e. Disconnect the video inputs and adjust the video gain and brightness controls on the front panel to obtain a blank raster.

f. Generate a red raster by turning off the green (S392) and blue (S393) screen switches inside the drawer.

g. Loosen the deflection yoke clamps and slide the yoke backward until a circular illuminated red area is displayed in the center of the screen.

h. Using the adjustment tabs, rotate the purity ring magnets, located on the convergence yoke, to obtain a pure red color over the entire circular area.

i. Slide the deflection yoke forward to its normal position and determine that the entire screen is illuminated uniformly with a pure red color. The purity ring magnets may be readjusted slightly to achieve uniformity. Tighten the deflection yoke into position.

j. Individually activate the green screen switch S392 and blue screen switch S393 to confirm that uniform green and blue rasters appear. If pure uniform rasters are not achieved, repeat the purity process steps f through i.

k. Following the purity alignment, it may be necessary to center the raster. Refer to the horizontal centering and vertical centering procedures in paragraphs 4.3.4.2 and 4.3.5.2.

4.3.8 Static Convergence Procedures

The red, green and blue convergence magnets are located on the ends of the Y-shaped convergence yoke. The thumbscrew adjustment for each beam may be rotated continuously in either direction. To obtain convergence correction, rotate the correction magnet pole axis. Each magnet moves its associated beam approximately along the direction from the CRT axis to the magnet position. The blue magnet moves the beam vertically and the red and green magnets move their beams at $\pm$ 60 degrees from the vertical. The blue lateral assembly, located at the rear of the convergence yoke assembly, is adjusted by a thumbscrew at the bottom of the assembly. The lateral adjustment moves all beams in a horizontal direction. The blue beam moves in the opposite direction from the red and green beams. The ratio of blue beam movement to red or green beam movement is five to one. To adjust the monitor, perform the following:

a. Position the monitor so that it faces a north or south direction. This will assure minimum effects from the earth's magnetic field.

b. If color impurities exist, refer to 4.3.7 for adjustment.

c. Apply a cross-hatch pattern to the video inputs.

d. Check to see if horizontal hold, focus, vertical linearity, and height are adjusted correctly.

e. Set the brightness to the desired level.

f. Adjust the red, green and blue center convergence magnets and the blue lateral magnet assembly to align the beams at the center of the CRT.

4.3.9 Pincushion Correction Circuit (PCC)

All pincushion adjustments are located on the control circuit card. Adjust PCC as follows:

a. Apply a composite cross-hatch video signal to the RGB video inputs.

b. Adjust the T/B PCC Amp Control VR704 to the center of rotation.

c. Adjust T/B PCC balance control VR703 for a straight horizontal line at the CRT center.

d. Adjust T/B PCC phase VR702 for symmetry of the cross-hatch lines at the right and left of screen.

e. Observe horizontal lines at the top of screen and adjust VR704 for a straight line.

f. Repeat steps d and e if necessary.

g. Adjust VR701 for a straight line at the right and left sides of the screen.

4.3.10 Screen Bias and Video Drive

4.3.10.1 Screen Bias

All screen adjustment controls are located behind the front panel. To adjust screen bias perform the following:

a. Turn on screen switches SW391, SW392, and SW393.

b. Apply video containing a cross-hatch pattern to the three video inputs.

c. Turn the ganged gain controls VR171, VR172 and VR173 and the brightness control VR691 fully counterclockwise.

d. Using a high-impedance probe, set voltages at terminals 13, 5 and 4, located on the CRT circuit card, to 500V by adjusting VR391, VR392 and VR393, respectively.

e. Adjust the brightness control VR691 to obtain a dim raster.

f. Observe the raster color to determine which CRT beams are not visible. Adjust the CRT screen voltage potentiometers (VR391 red, VR392 green and VR393 blue) as required to equalize the three beam intensities. This adjustment will result in a white raster.

4.3.10.2 Video Drive

To adjust video drive, proceed as follows:

a. Prior to performing the video drive adjustment, the screen bias adjustments must be adjusted properly.

b. Set the brightness control VR691 to produce a blank (dark) screen.

c. Set the three contrast controls VR174, VR175 and VR176, located in the drawer, to the center of their ranges.

d. Adjust the ganged gain controls (VR171, VR172 and VR173) to obtain maximum useable highlight brightness.

e. Observe the highlight color and adjust the three contrast controls VR174, VR175 and VR176 to obtain white highlights.

4.3.11 Focus

Focus can be attained by performing the following:

a. Adjust the brightness and gain controls located on the front panel for a normal display.

b. Adjust the focus control VR591 located on the HV assembly for best overall focus, observing both the center and corners of the screen.

c. Adjust dynamic focus VR501 located on the control circuit card to obtain uniform focus.

d. Repeat steps b and c if necessary.

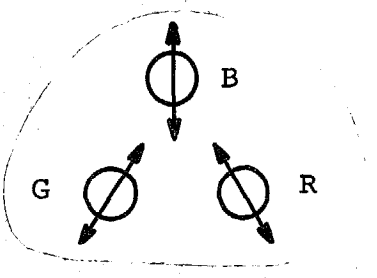
4.3.12 Dynamic Convergence Alignment Procedure

To adjust dynamic convergence, proceed as follows:

a. Converge the center of the screen using the static convergence procedure. As the dynamic convergence procedure is performed, the center convergence should be checked and readjusted as necessary.

b. Make sure that purity, height, width, vertical linearity and focus have been properly adjusted before performing next step.

c. The numbers next to each control indicate the order in which the controls are to be adjusted and the corresponding screen area to be effected. All ten RGB potentiometer groups give the following beam movements:



Each group of controls has minimal interaction.

d. After all ten potentiometer groups have been adjusted, any group may be readjusted for small amounts of additional correction.

4.3.13 Video Circuits

4.3.13.1 Pedestal Level

To adjust pedestal level, perform the following:

a. Apply a composite video signal of 1V P-P to the green video input.

b. Observe the green video drive at the green terminal on the video card, using a dc-coupled oscilloscope.

c. Set the contrast and the gain control at the center of rotation.

d. Using a dc oscilloscope, adjust clamp level control VR271, located on the video circuit card, to obtain a pedestal level of 100V $\pm$ 1V at terminal 6 on the CRT socket.

4.3.13.2 Video Frequency Response

To adjust video frequency response, proceed as follows:

a. Apply a window pattern of .5 to 2V P-P or a square wave signal of about 50 KHz to the blue video input. This test pattern should have rise and fall times of 30 nanoseconds or less.

b. Using a dc oscilloscope, observe the output waveform at terminal 11 on the CRT socket.

c. Adjust capacitor C103 to minimize video pulse overshoot or undershoot. The oscilloscope probe must be properly equalized prior to making this adjustment.

d. Follow the same procedure for the red and green channels.

red - terminal 2 - Capacitor C113
green - terminal 6 - Capacitor C123

4.3.13.3 Video Peaking

To achieve video peaking, perform the following:

a. Apply a composite video signal, containing a frequency sweep which covers a frequency range of from 500 kHz to 15 MHz, to each video input.

b. Connect the oscilloscope high-impedance (10 to 1) probe, through a two picofarad capacitor, to the blue CRT cathode terminal (pin 11 of the CRT socket).

c. Adjust the input video level to obtain a low frequency output level of 30V P-P.

d. Adjust coils L201 and L202 to obtain a flat frequency response ranging up to 15 MHz.

e. Following the above procedure, adjust the frequency response of the red and green channels.

red - terminal 2 - Coils L211, L212
green - terminal 6 - Coils L221, L222

4.3.14 Automatic Beam Limiter (ABL) Adjustment

To make ABL adjustment, perform the following:

a. Apply composite video signals to the input terminals.

b. Connect the probe of a Vacuum Tube Voltmeter (V.T.V.M.) to test point 1 on the H-V circuit board. Set the scale of the V.T.V.M. to measure -15V.

c. Adjust potentiometer VR601, located on the video circuit card, the contrast and the gain controls fully counterclockwise. Then turn the brightness control full clockwise.

d. Adjust VR601 to obtain -13.2 $\pm$ .5V on the V.T.V.M.

SECTION 5

PARTS

The following table contains a list of replaceable subassemblies and chassis piece parts. In Table 5-1, the subassemblies are listed first, followed by the chassis piece parts. An Aydin Controls part number is included for each item. The last column indicates those items which are recommended spare parts.

Reference Number	Aydin Controls Part Number	Description	Recommended Spares
	195-460-023	PCB Convergence	*
	195-460-024	PCB Control	*
	195-460-025	PCB Power-Regulator	*
	195-460-026	PCB Video	*
	195-460-027	PCB Power-Rectifier	*
	195-460-028	PCB Connector	
	195-460-029	PCB CRT	*
	195-460-030	PCB Deflection	*
	195-460-031	PCB High Voltage	*
	195-105-007	Deflection Yoke	
	195-105-006	Convergence Yoke	
	195-109-009	Transformer FBT	
	195-109-010	Transformer Power	
	195-136-002	Focus Pack	
	195-105-005	Tube Cathode Ray 19 Inch	
	195-103-003	Transistor 2SC 1413	
	195-103-001	Transistor 2SC 1106	
	195-103-004	Transistor 2SC 1586	
	195-107-006	Switch S191-S194	
	195-107-007	Sw. slide S391-S393	
	195-107-008	Sw. slide S171	
	195-107-009	Sw. Rocker S992	
	195-107-010	Sw. Rocker S991	
	195-101-008	VR391-VR393	
	195-101-009	VR174-VR176	
	195-101-010	VR171-VR173	
	195-101-011	VR471	
	195-101-012	VR571	
	195-101-013	VR691	

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